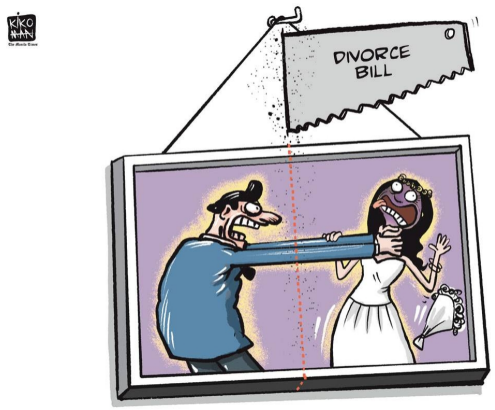
# Difference-in-Differences

## Staggered Adoption: Three Parallel-Trends Assumptions, Three Estimators

F. Candau (UPPA-TREE)

# Staggered adoption

- In chapter 4, the treatment starts at **one date** for all treated units.
- Most policies **roll out**: units adopt at different dates, and some never adopt.
- Running example: **unilateral divorce** in the US. From 1970, states let one spouse end the marriage without the other's consent. Most states adopt in the 1970s, some never do.
- Stevenson and Wolfers (2006) estimate its effect on **women's suicide**.




*Reference for the whole chapter: Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (2026), "Difference-in-Differences Designs: A Practitioner's Guide".*

# Roadmap

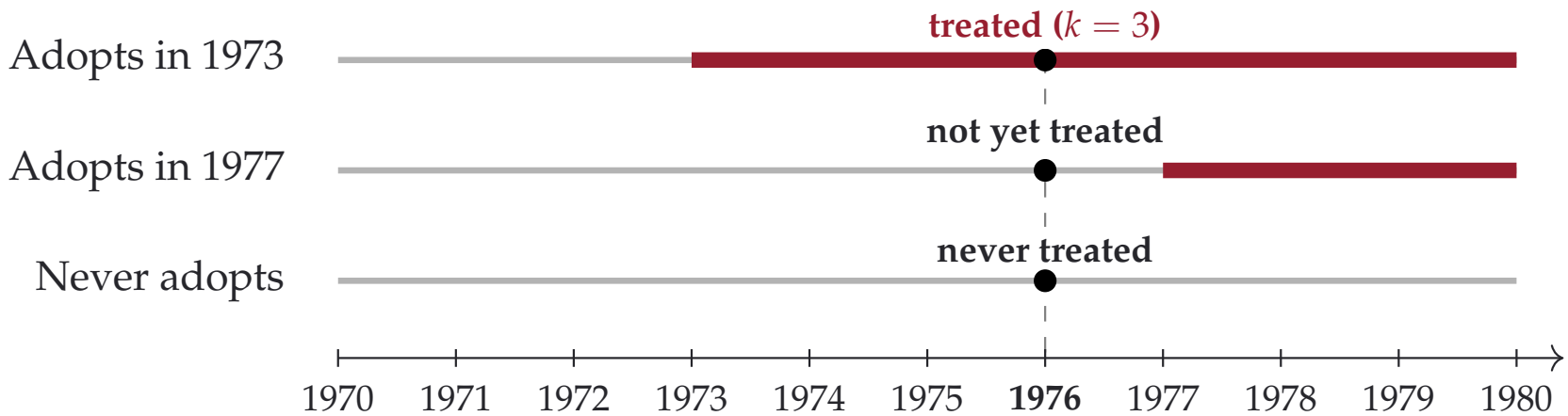
1. **Setting**: cohorts and comparison groups.
2. **The problem**: what TWFE computes when adoption is staggered.
3. **The solution**: one $2 \times 2$ per cohort and date, three comparison groups, three estimators.
4. **Aggregation** and conclusion.

Section 1

# Setting

# Cohorts and comparison groups

- **Cohort** $g_i$: the date at which unit $i$ is first treated ($g_i = \infty$ if never treated).
- **Event time** $k = t - g_i$: the number of periods since adoption.
- At date $t$, a unit is **treated**, **not yet treated** or **never treated**. In 1976:




- Not-yet-treated and never-treated units are the two possible **comparison groups**.

Section 2

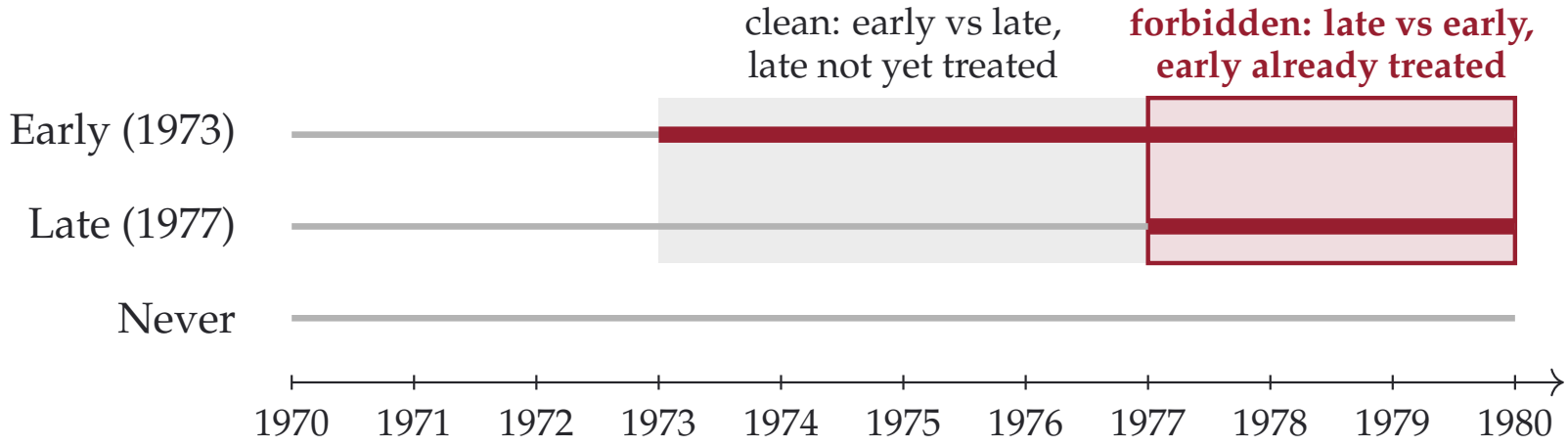
# TWFE

# The static TWFE under staggered adoption

$$y_{it} = f_i + f_t + \beta\, d_{it} + u_{it}, \qquad d_{it} = \mathbf{1}\{t \geq g_i\}$$

- $\hat{\beta}$ is a weighted average of **all** the $2 \times 2$ DiDs in the data (Goodman-Bacon 2021).
- With an early cohort, a late cohort and never-treated units, there are four $2 \times 2$s:
  - early cohort against never-treated units;
  - late cohort against never-treated units;
  - early cohort against the late cohort, **before** the late cohort is treated;
  - late cohort against the early cohort, **already treated**: a **forbidden comparison**.
- The weights depend on group sizes and on the timing of adoption.

# The forbidden comparison




- From 1977, TWFE compares the late cohort with the early cohort, **already treated**.
- The early cohort's outcome still moves because of its own treatment: its change is not a counterfactual.
- TWFE subtracts this change. The early cohort's effect receives a **negative weight** (de Chaisemartin and D'Haultfœuille 2020), and $\hat{\beta}$ can shrink or even change sign.
- Divorce: TWFE gives $-3.1$ suicides per million women, $-5.4$ without the forbidden comparisons (Goodman-Bacon 2021).

# The dynamic TWFE

$$y_{it} = f_i + f_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - g_i = k\} + u_{it}$$

- The event-study regression has the same problem (Sun and Abraham 2021).
- Each $\hat{\beta}_k$ mixes the effect $k$ periods after adoption with the effects of **other cohorts at other dates**.
- The early cohorts' effects can appear in the **pre-period** coefficients, as if parallel trends failed.
- The problem disappears only if all cohorts have the same effect.

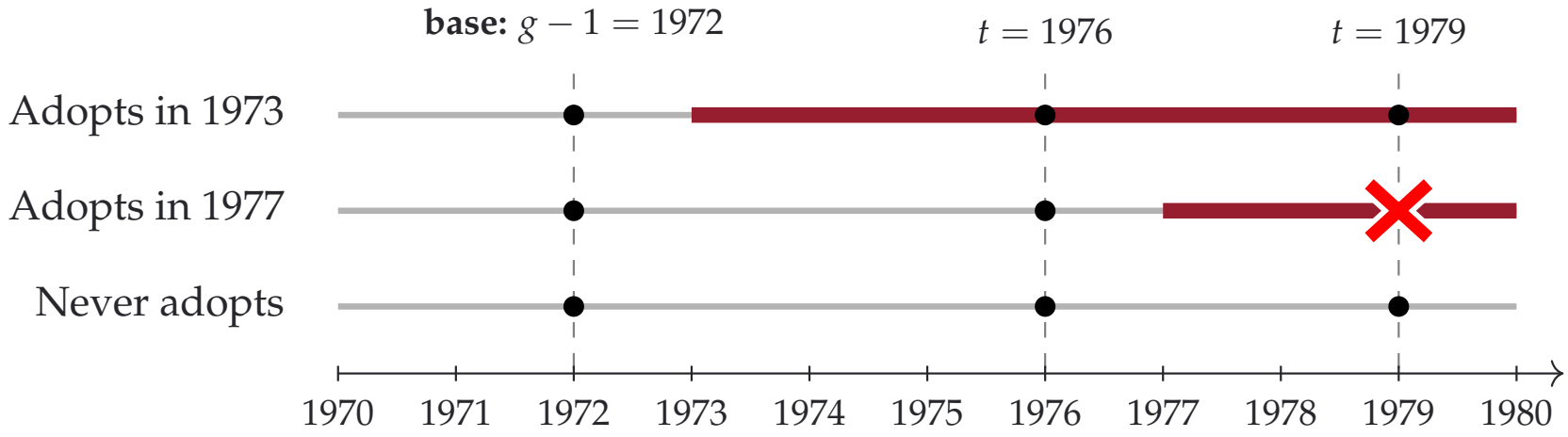
Section 3

# Solution

# The solution: choose the comparison group

- TWFE mixes **all** the $2 \times 2$s, forbidden ones included.
- Instead, estimate $\text{ATT}(g, t)$, the effect at date $t$ for the units first treated at $g$, against a comparison group **chosen** by the researcher.
- Choosing the comparison group is choosing a parallel-trends assumption.
- The $\text{ATT}(g, t)$ are then averaged.

# Example: the comparison group for ATT$(1973, t)$




- The effect in 1976 for the states that adopted in 1973: their change from 1972 to 1976, minus the change of the comparison group over the same years.
- Comparison group = **never-treated**: the states that never adopt.
- Comparison group = **not-yet-treated**: the states that never adopt **and** the states that adopt in 1977, still untreated in 1976.
- For a later date, say 1979, the states that adopt in 1977 are already treated and leave the not-yet-treated group: its composition changes with $t$.

# Three parallel-trends assumptions

| Assumption | Comparison group | Base period |
|---|---|---|
| **PT-never** | never-treated units | $g-1$ |
| **PT-not-yet** | units not yet treated at $t$, never-treated included | $g-1$ |
| **PT-all** | not-yet-treated units | all pre-periods |

- PT-never and PT-not-yet assume parallel trends only **after** $g-1$.
- PT-all also assumes them **before**, so the pre-trends become a test of the assumption.

# Three estimators

| Estimator | Assumption | How it works |
|---|---|---|
| Callaway and Sant'Anna (2021) | PT-never or PT-not-yet | computes each ATT$(g,t)$ as a $2 \times 2$, then averages them |
| Sun and Abraham (2021) | PT-never | the same effects, from a regression with one dummy per cohort and event time |
| Imputation (Borusyak et al. 2024; Gardner 2022) | PT-all | estimates the fixed effects on untreated observations, then predicts the untreated outcome of treated ones |

*Other estimators: stacked regressions (Cengiz et al. 2019); de Chaisemartin and D'Haultfœuille (2020), which also handle treatments that switch off (chapter 6).*

# Choosing an assumption

- **PT-all** (imputation): the most **precise**, but biased if the pre-trends are not parallel.
- **PT-never**: a comparison group of **fixed composition**, but never-treated units may be **too few** or **too different**.
- **PT-not-yet**: more comparison units than PT-never, but their composition **changes** with $t$.

## Not a robustness check

The estimators rest on different assumptions, so their estimates differ. Report the estimator of the assumption you can defend; a gap with the others shows which assumption the result depends on.

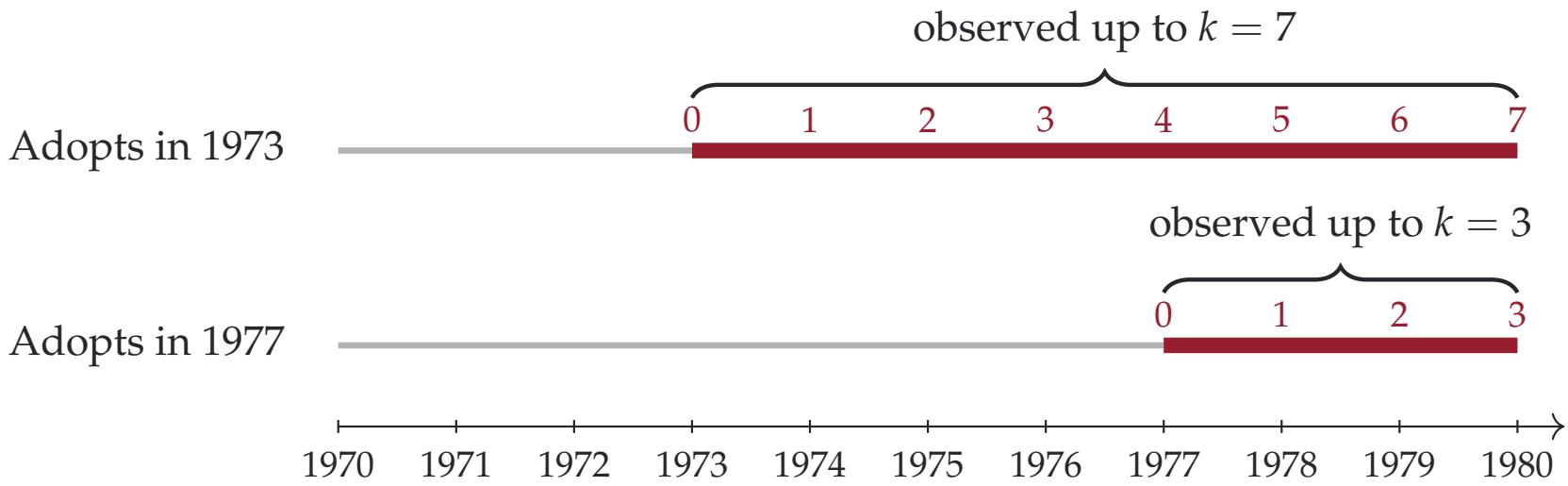
Section 4

# Aggregation

# Aggregation

- Each ATT$(g, t)$ compares a **treated cohort**, the units first treated at $g$, with a **comparison group**, never-treated or not-yet-treated units.
- Forbidden comparisons concern the comparison group. **Aggregation concerns the treated cohorts.**
- The event study averages, for each $k$, the effects **of the treated cohorts** observed $k$ years after adoption.
- The late cohorts are observed **for fewer years** after adoption, so the cohorts in the average **change with** $k$.

# Example: unequal windows




- Data stop in 1980: the 1973 states are observed up to $k = 7$, the 1977 states only up to $k = 3$.
- From $k = 0$ to $k = 3$, the event study averages the two cohorts. **From $k = 4$ on, it covers only the 1973 states.**
- A jump between $k = 3$ and $k = 4$ can therefore come from the 1977 states leaving the average, with no change in the effect.
- Either keep a window common to all cohorts ($k \leq 3$ here), or say it.

# Conclusion

1. **Define the target**: which effects, which average.
2. **Choose the comparison group**, that is the parallel-trends assumption, and defend it.
3. **Estimate** with the corresponding estimator.
4. **Check** the pre-trends and compare with the neighbouring assumption.

## On TWFE

"It has well-understood, potentially serious, and easily remedied problems, and we do not recommend using it." — Baker et al. (2026)

Section 5

# In R

# What you must build in the data

These estimators need the **timing** structure explicitly:

- the **cohort** $g_i$ — each unit's first treatment period, on *all* of its rows;
- **event time** $k = t - g_i$;
- a **never-treated** indicator (or the last-treated cohort as never-treated, dropping the periods where everyone is treated);
- units treated in the **first** period have no pre-period: drop them.

Software builds the $2 \times 2$s and the aggregation for you — you supply $g_i$ and the **assumption**.

# In R: the timing variables (data.table)

Six lines build the variables the authors' R packages need, with `data.table`.

```
library(data.table); library(fixest); library(did)
library(did2s); library(HonestDiD)
d <- fread("rica_bio_td5.csv")                                          # 1
d[, e := fifelse(cohorte > 0, annee - cohorte, NA_integer_)]            # 2
d[, traite := as.integer(cohorte > 0 & annee >= cohorte)]               # 3
d[, ee := e]; d[ee < -4 | ee > 5 | is.na(ee), ee := -1000L]             # 4
d[, g_sa := fifelse(cohorte == 0, 10000L, cohorte)]                     # 5
d[, reg_an := paste(region, annee)]                                     # 6
```

- **1** — one row per farm `id` and year `annee`; `cohorte` is $g_i$, the conversion year, 0 if never converted; `y` on the next slides stands for the outcome column.
- **2** — event time $k = t - g_i$, missing if never converted (`d[, x := ...]` adds a column; `fifelse` is a vectorised if).
- **3** — $d_{it} = \mathbf{1}\{t \geq g_i\}$, the treatment indicator of TWFE.
- **4** — event time **binned**: outside $[-4, 5]$ and never converted get a reference code.
- **5** — the cohort as `sunab` wants it: never converted coded as a far-future cohort.
- **6** — the region $\times$ year label of the fixed effect.

# In R: TWFE and Sun–Abraham (fixest)

Four lines estimate the regression to avoid and its saturated correction.

```
tw <- feols(y ~ traite | id + reg_an, d, cluster = ~id)          # 1
sa <- feols(y ~ sunab(g_sa, annee, ref.p = -1) | id + reg_an,
            d, cluster = ~id)                                    # 2
summary(sa, agg = "ATT")                                         # 3
coeftable(sa)                                                    # 4
```

- **1** — TWFE: the outcome on $d_{it}$ with farm and region × year effects, standard errors clustered by farm. Its coefficient is the weighted average of 2 × 2s of the chapter, forbidden comparisons included.
- **2** — Sun–Abraham: `sunab(cohort, period)` builds every cohort × event-time dummy, with $k = -1$ as reference; the never converted are the only control.
- **3** — aggregates the cohort × time coefficients into one ATT with the cohort shares as weights.
- **4** — the event-study coefficients, one per $k$, with their standard errors.

# In R: imputation (did2s)

Two lines estimate the PT-all analogue and test its pre-trends.

```
imp <- did2s(d, yname = "y", first_stage = ~ 0 | id + reg_an,
             second_stage = ~ i(ee, ref = c(-1, -1000)),
             treatment = "traite", cluster_var = "id")          # 1
wald(imp, keep = "ee::-[2-4]$")                                 # 2
```

- **1** — imputation in two stages. `first_stage`: the fixed effects are estimated on the **untreated cells only**, those where `traite` is 0. `second_stage`: the outcome net of those effects is regressed on the event-time dummies, `i(ee, ...)`, with $k = -1$ and the binned code as references. Standard errors clustered by farm.
- **2** — the joint test that the coefficients of $k = -4, -3, -2$ are zero. fixest names the dummies `ee::-4`, `ee::-3`, ...; `keep` selects them by a pattern on the name. This is the pre-trend check of PT-all.

# In R: Callaway–Sant'Anna (did)

Two lines estimate every ATT$(g,t)$ and aggregate them by event time.

```
a <- att_gt(yname = "y", tname = "annee", idname = "id",
            gname = "cohorte", data = d,
            control_group = "notyettreated",
            base_period = "universal",
            allow_unbalanced_panel = TRUE)                    # 1
aggte(a, type = "dynamic", min_e = -4, max_e = 5)             # 2
```

- **1** — one ATT$(g,t)$ per cohort and year, each a $2 \times 2$ against the **not-yet-treated**; `"nevertreated"` gives the other assumption. `"universal"` takes $g-1$ as the base period for the pre-periods too (the default compares each pre-period with the previous one). On a rotating panel, `allow_unbalanced_panel` makes the package treat the data as repeated cross-sections.
- **2** — aggregates the ATT$(g,t)$ by event time, from $-4$ to 5; `type = "simple"` returns a single ATT instead.

# In R: how much violation survives (HonestDiD)

Three lines bound the effect at one event time under a limited violation of parallel trends.

```
beta <- coef(imp); V <- vcov(imp)                                    # 1
hd <- createSensitivityResults_relativeMagnitudes(
        betahat = beta, sigma = V, numPrePeriods = 3,
        numPostPeriods = 6, Mbarvec = c(0.5, 1, 1.5, 2),
        l_vec = basisVector(4, 6))                                   # 2
hd[, c("Mbar", "lb", "ub")]                                          # 3
```

- **1** — the event-study coefficients of the imputation and their covariance, ordered from $k = -4$ to $k = 5$: three pre-periods $(-4, -3, -2)$ and six post $(0, \ldots, 5)$; check the order with `names(beta)`.
- **2** — Rambachan–Roth: for each $\bar{M}$, the interval that stays valid if the post-treatment deviation from parallel trends is at most $\bar{M}$ times the largest pre-treatment one; `l_vec` selects the fourth post period, $k = 3$.
- **3** — one row per $\bar{M}$ with the bounds. The first $\bar{M}$ whose interval contains zero measures how thin the margin is.

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